

Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 7A nr. 16.4

$$f(x, y) = \begin{cases} \frac{x^2 y^8}{x^6 + y^{12}} & \text{als } (x, y) \neq (0, 0) \\ 0 & \text{als } (x, y) = (0, 0) \end{cases}$$

Continuïteit:

Stel $x = y^2$

$$\begin{aligned} \lim_{y \rightarrow 0} f(y^2, y) &= \frac{y^{12}}{y^{12} + y^{12}} \\ &= \frac{1}{2} \end{aligned}$$

Nu voor $x = 0$

$$\begin{aligned} \lim_{y \rightarrow 0} f(0, y) &= \frac{0y^8}{0 + y^{12}} \\ &= 0 \end{aligned}$$

$\Rightarrow f$ is niet continu in $(0, 0)$

Partiele afgeleiden:

$$\begin{aligned} \frac{\partial f}{\partial x}(0, 0) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda, 0) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda^6} = 0 \\ \frac{\partial f}{\partial y}(0, 0) &= \lim_{\lambda \rightarrow 0} \frac{f(0, \lambda) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda^{12}} = 0 \end{aligned}$$

$\Rightarrow f$ is partieel afleidbaar in $(0, 0)$

Afleidbaarheid:

Stel $\bar{h} = (h_1, h_2)$, $h_1 \neq 0 \neq h_2$

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda h_1, \lambda h_2) - f(0, 0)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\frac{\lambda^{10} h_1^2 h_2^8}{\lambda^6 h_1^6 + \lambda^{12} h_2^{12}}}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^{10} h_1^2 h_2^8}{\lambda^7 (h_1^6 + \lambda^6 h_2^{12})} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^3 h_1^2 h_2^8}{h_1^6 + \lambda^6 h_2^{12}} \end{aligned}$$

$$= 0$$

$\Rightarrow f$ is afleidbaar in $(0, 0)$

Differentieerbaarheid:

f is niet continu in $(0, 0)$

$\Rightarrow f$ is niet differentieerbaar in $(0, 0)$

References